

Voltammetry

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Objectives

- **After studying this session you should be able to:**
 - Define polarography,
 - Discuss the details of a dropping mercury electrode,
 - Discuss the limiting and migration currents,
 - Discuss the diffusion and residual currents,
 - Discuss the factors affecting the diffusion current,
 - Discuss the instrument, and
 - Discuss the details of amperometric titrations

Introduction

- ▶ Voltammetry is the general name given to a group of **electroanalytical methods** in which the *current is measured as a function of applied potential*.
- ▶ Time dependent potential is applied to an **electrochemical** cell, and the **current flowing** through the cell is measured as a **function of that potential**.
- ▶ A plot of current as a function of applied potential is called a **voltammogram**.

History



Jaroslav Heyrovsky

"for his discovery and development of the polarographic methods of analysis".

(1959)

- The **polarographic** method of analysis was developed by Jaroslav Heyrovsky in 1922 which is the earliest **voltammetric technique introduced**.
- He received Nobel Prize in 1952 for developing this technique using **dropping mercury electrode** as the working electrode.
- He called the recorded current-potential curves as **polarograms** and introduced the term **polarography**.

Principle of voltammetry

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- ➔ When a **potential difference** is applied across the **working, reference** and **auxiliary electrodes** that immersed in the **sample solution** containing a **supporting electrolyte** and the **analyte**(electroactive compound);
 - A small **current arises** due to background reactions and dissolved impurities.

- **The reduction process taking place at the electrode.**
- The reduction at the electrode increases with the voltage applied on electrodes & the current is recorded.
 - The metallic ions are reduced at the surface of the electrode.
 - Migration of the ions from the solution to the electrode surface

Instrumentation

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Instrumentation – Three electrodes in solution containing analyte

Working electrode: microelectrode whose potential is varied with time

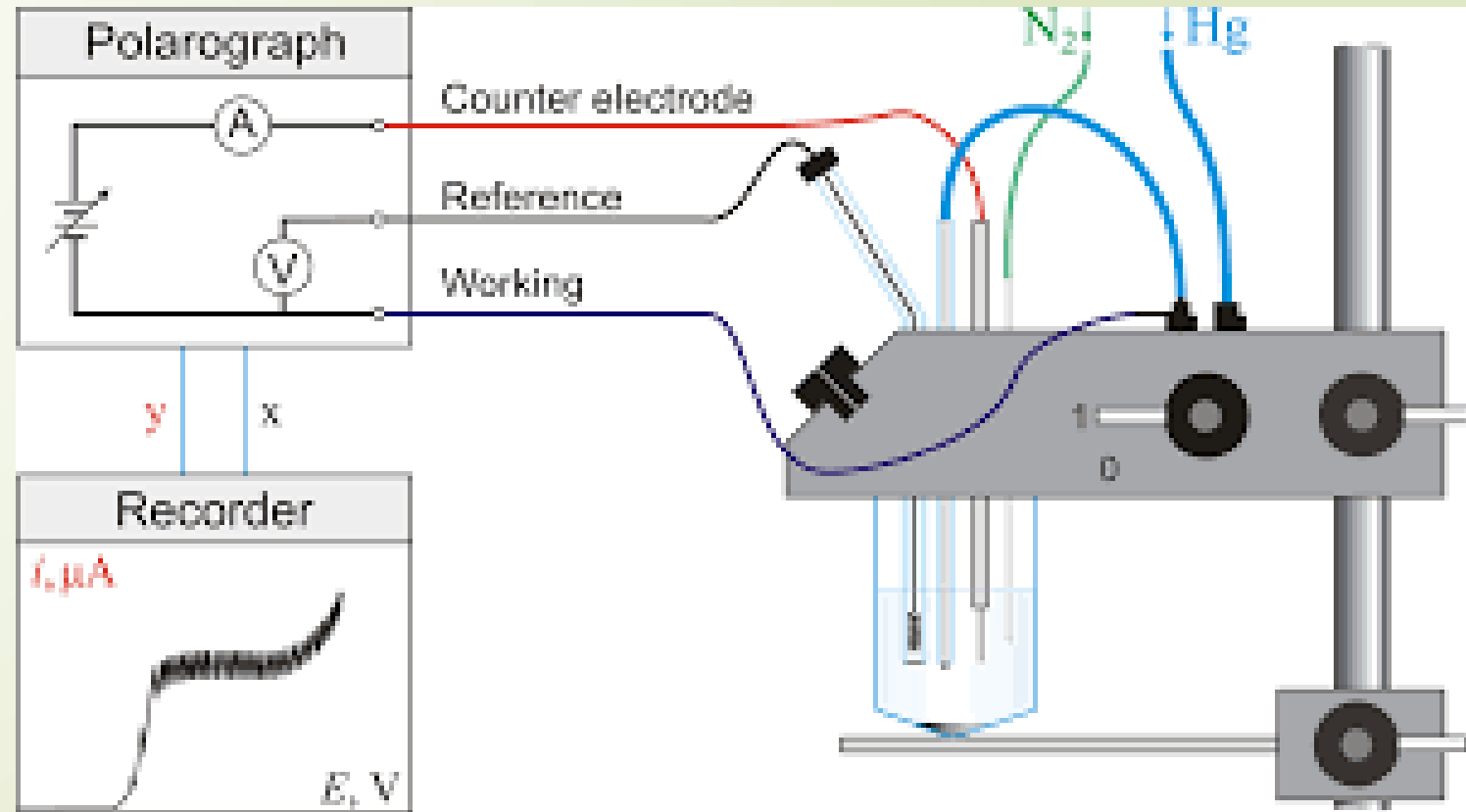
Reference electrode: potential remains constant (Ag/AgCl electrode or calomel)

Counter electrode: **Hg** or **Pt** that completes circuit, conducts e^- from signal source through solution to the working electrode.

Supporting electrolyte: excess of nonreactive electrolyte (alkali metal) to conduct current.

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Voltammetry.....

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➤ **Voltammetry** comprises a group of electro-analytical methods in which **information about the analyte** is obtained by *measuring current as function of applied potential* either **variable or static**:

i. When current proportional to analyte concentration is monitored at **fixed potential**, the technique is called **Amperometry**.

ii. When current proportional to analyte concentration is monitored at **variable potential**, the technique is: polarography.

Polarography

- The first important voltammetry technique to be developed.
 - **Uses the dropping mercury electrode as the working electrode.**
 - Since mercury is a liquid, the working electrode often consists of a drop suspended from the end of a capillary tube as a result of gravity.

- The drop forming at the capillary **tip grows**, and **it tears away** when at a time after the start of formation it has attained a certain mass.
- After its detachment ***a new drop starts to form and grow***, and the cycle is repeated.
- **The growing drop, while suspended, is used as the cathode** for reduction of various substances.

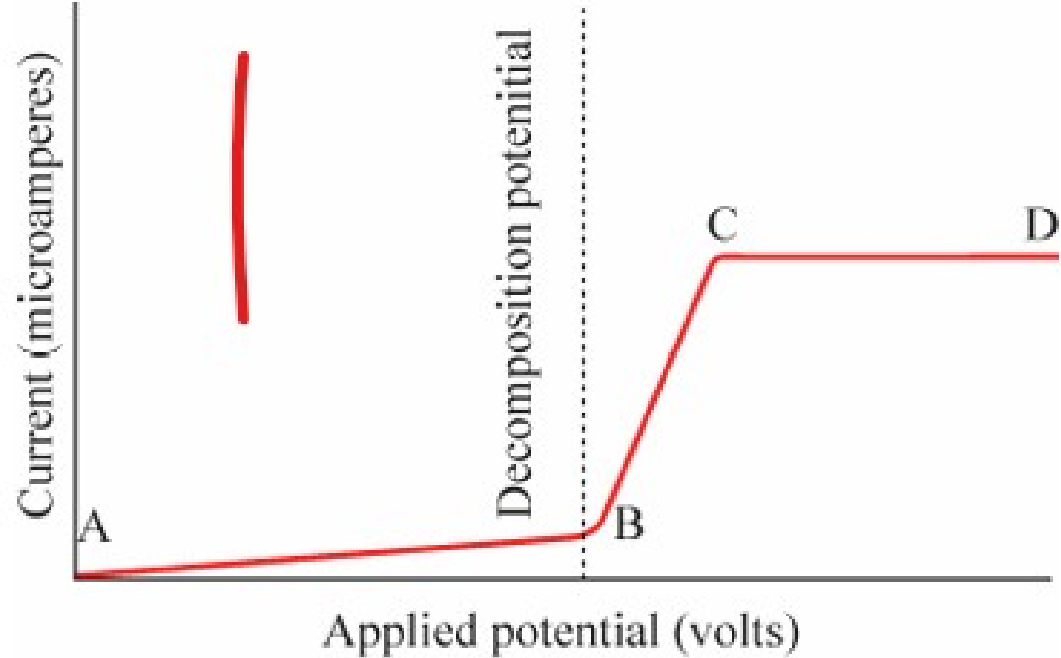
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according to Ilkovic equation, i_d is proportional to concentration keeping all other factors of the equation constant.

- So, if some of the electroactive material in the solution is removed by interaction with some other reagent (e.g. EDTA reagent for Zn^{2+} determination) the diffusion current will decrease.
- This is the fundamental principle of amperometric titrations or polarographic titrations.

During electrolysis, chemical reactions occur under **the influence of an electromotive force** at electrodes immersed in solutions.

- The reactions occurring through the **transfer of electrons** between the **electrode** and the **substances in the solution**.
- Because of this the **equality of positive and negative ions** are **unbalanced** at the electrode but is counter acted by oppositely directed movement of positive and negative ions through the solution to **produce the flow of current** through the solution.



- When the applied potential is equal to the decomposition potential (B) of the electroactive substance at the cathode the current starts.

Typical polarogram

- This is the current which is actually measured by the movement of electrons in the external circuit connecting the two electrodes.
- Therefore, the current is the result of the electrode reaction.**

Advantages DME

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➔ Its fluid nature **eliminates two problems**

I. Contamination of the electrode surface resulting from electrode reactions

- The mercury surface is continuously renewed in the case of the **DME.**

II. The decay of the current to small values with time due to *limited transportation of analyte to the electrode surface when the solution is not stirred*

- The solution is stirred in the case of the DME each time the

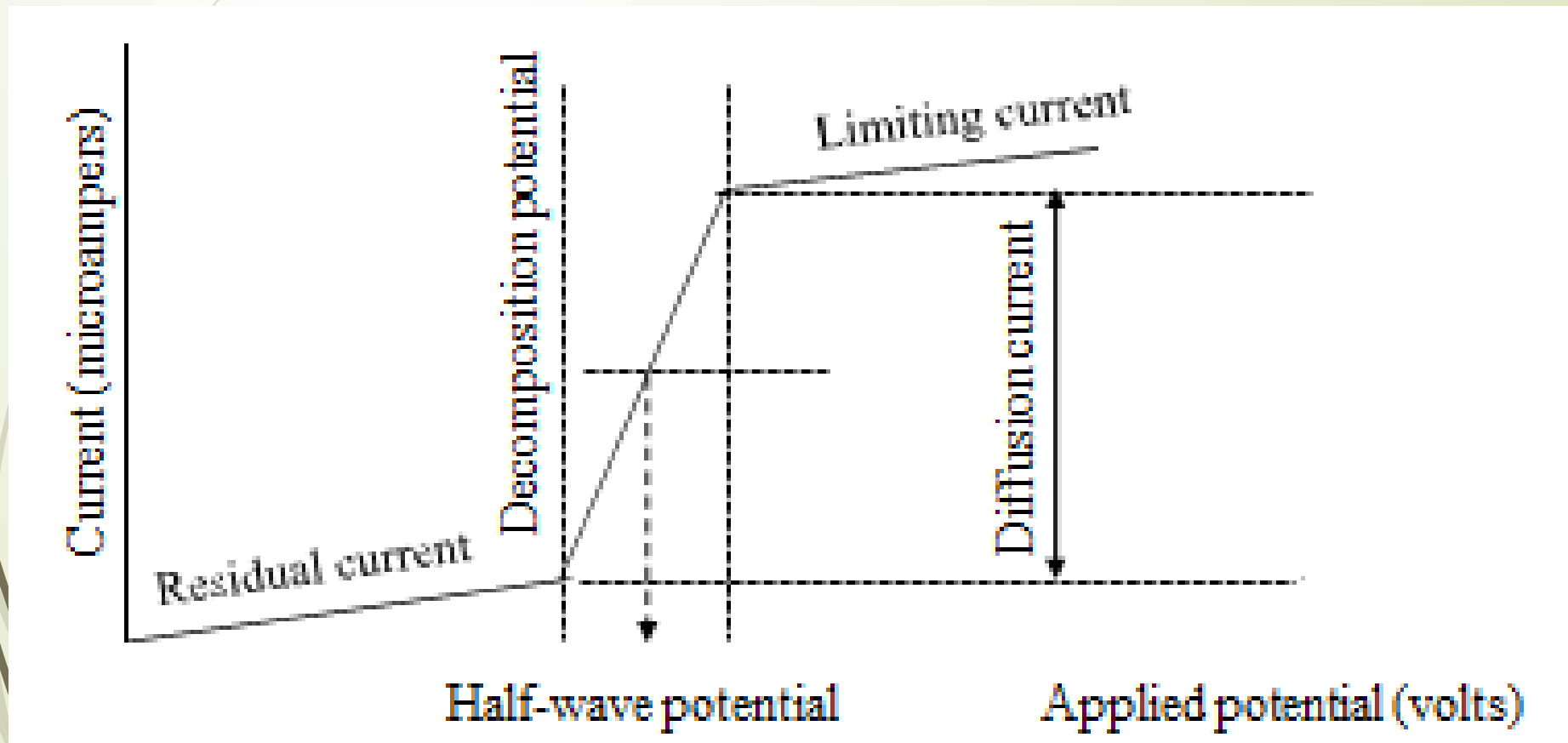
Disadvantages DME

- ✓ The capillary is very small in size so **easy to block** and so all process stopped.
- ✓ The presence of **impurities** in the sample or in mercury cause **blockage of the capillary** results in malfunction of the electrode.

Polarogram

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- ➔ A plot of **Current vs voltage** - polarogram.



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- **Current** is measured in polarography
- **Parameters** associated with the polarogram are:
 - **Diffusion current (i_d),**
 - **Half-wave potential ($E_{1/2}$),**
 - **Back ground (Residual) current,**
 - **The limiting current and**
 - **Decomposition potential**

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1-Residual current: at small applied potential only a residual current flow in the cell which is caused by: The **presence of impurities** in the supporting electrolyte or sample.

2-Limiting current: is the region on the polarogram in which the current after increased sharply , becomes essentially *independent on the applied potential* .

3-Diffusion current: it is the difference between limiting and residual currents .

☞ It is **directly proportional to the concentration of the reactive substance.**

☞ Ilkovic examined the various factors which govern the diffusion current obtained with DME.

4. Half wave potential ($E_{1/2}$) : is the potential at which the **current is equal to one half the diffusion current.**

► Factors affecting half wave potential

- A characteristic of the nature of the reactive material
- Independent on concentration
- Type of electrode.

5- Decomposition potential:

is the potential at which the **substance begin to oxidize or reduce.**

Ilkovic Equation:

- Ilkovic equation is a relation used in polarography relating the **diffusion current (i_d)** and **the concentration**.

$$I_d = 607nD^{1/2}cm^{2/3}t^{1/6}$$

Where

I_d - **diffusion current (micro-Amper)**,

C-concentration (M)

D- diffusion coefficient (cm²/seconds),

m-rate of flow of the mercury and

t- drop time (seconds)

n-number of electrons transferred

Factors affecting the diffusion current

- The ilkovic equation shows that there is a **linear relationship** between **id** and **concentration of electroactive substance**.
- It also accounts for **many factors** which **influence the diffusion current and these factors are**,
 - The concentration of electroactive substance, the dimensions of the capillary and the pressure on the dropping mercury.

Factors cont....

- Temperature, the viscosity of the medium, the composition of the base electrolyte etc..

1. The Concentration of Electroactive Substance:

- According to the Ilkovic equation

$$I_d = kc$$

$$I_d/c = k$$

is constant keeping all other factors constant.

- This is the basis of quantitative polarographic analysis.

2. The Capillary Characteristics and the Factors Affecting Them

- ➡ From Ilkovic equation diffusion current depends on 'm' and 't' (together referred as **capillary characteristics**).
- ➡ and are therefore affected by their variations.
- ➡ 'm' is directly proportional and 't' is inversely proportional to the **net pressure on the mercury drop**.

3. Temperature

- The change in temperature affects all values in the Ilkovic equation except the value of 'n'.
- The temperature coefficient of 'm' is about +0.27% per degree, while that of 't' is about – 0.22% per degree.
- So the temperature coefficient $m^{2/3} t^{1/6}$ is about +0.15% per degree.

- In addition, the numerical constants 607 is also affected and the temperature coefficient of this constant K is +0.00012 per degree.
- Combining all these values gives the temperature coefficient of the diffusion current as +0.013 per degree.
- So the diffusion current increases about 1.3% per one degree rise in temperature.
- **Therefore, it is advisable to keep the temperature constant within + 0.5o .**

4. The Solvent and the Supporting Electrolyte

- The **diffusion current** depends on 'D' of the electroactive species which in turn depends on the **viscosity of the solution**.
- 'id' decreases with increase in the viscosity of the solution as **D is inversely proportional to the viscosity, η of the medium**.
- Changes in the composition of the supporting electrolyte by adding **complexing ligands** to the **electrolyte solution** changes the **size of the metal complex** and **therefore D of the ion**.

Cont....

- I_d therefore decreases with complexation.
- It is necessary to keep the concentration of every constituent of the supporting electrolyte constant within about $\pm 10\%$ to ensure the variation of i_d resulting from this effect (viscosity) will be **negligibly small**.

Polarographic equation

- Let us take a general electrode reaction, process of reduction at the cathode is accompanied by a simultaneous oxidation as free electrons cannot exist in solution.



- The electrode potential for such reversible reaction is given by

$$E = E^{\circ} + \frac{RT}{nf} \log \frac{(ox)}{(red)} \dots (2)$$

Cont..

- The current observed depends upon the rate of the diffusion of oxidant and is given by.

$$i = K_1 ([Ox] - [Ox]_0) \dots\dots\dots(3)$$

so that according to Ilkovic equation K_1 is equal to $607 n_{ox} D_{ox}^{1/2} m^{2/3} t^{1/6}$

- When the current attains the limiting value concentration of oxidant at the electrode-solution interface will be essentially zero.
 - as it is reduced as soon as it reached the interface.
- The current therefore is ***i_d*** and Eq. may be written as,

$$\rightarrow i = k_1(\text{OX}) = i_d \dots (4)$$

From eq 3&4

$$(\text{ox})_o = i_d - i / k_1$$

➔ If the **metal ion (oxidant)** is **reduced to metal (reductant)**

➔ Therefore, the concentration of reductant formed is proportional to the observed current, so

$$i = K_2[\text{Red}]_o$$

$$\text{where } K_2 = 607 n_{\text{red}} D_{\text{red}}^{1/2} m^{2/3} t^{1/6}$$

➔ Substituting these in Eq 2.

$$E = E^\circ - \frac{0.05916}{n} \log \frac{K_1}{K_2} - \frac{0.05916}{n} \log \frac{i}{i_d - i}$$

- ▶ when the potential of the dropping electrode is equal to the half-wave potential, $E_{1/2}$

where $i = i_d/2$ the Eq. then becomes,

$$E = E_{1/2} - \frac{0.05916}{n} \log \frac{i}{i_d - i}$$

- ▶ This equation is termed as “Equation of the Polarographic Wave” and represents the potential as a function of current at any point on the polarographic wave.

Exercise

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1. What do you mean by
 - a) polarography & Polarogram
2. What are the advantages of a DME?
3. What do you mean by residual current?
4. Distinguish between limiting current and diffusion current
5. What is referred to as capillary characteristics and what are the factors affecting them?
6. In what range should the temperature be kept during polarographic experiment and why?
7. What is the polarographic equation and what does it represent?

AMPEROMETRY

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- The **amperometric** measurements are mainly based on the **polarographic principles** & the **measurement of the current at a fixed potential**.
- *Diffusion current proportional to analyte conc.* (like in polarography) is monitored at a **fixed potential**.

Cont....

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- Because we **do not vary the potential**, amperometry does not result in a **voltammogram**.
- One important application of amperometry is in the construction of **chemical sensors**.
 - One of the first amperometric sensors to be developed was for dissolved O₂ in blood

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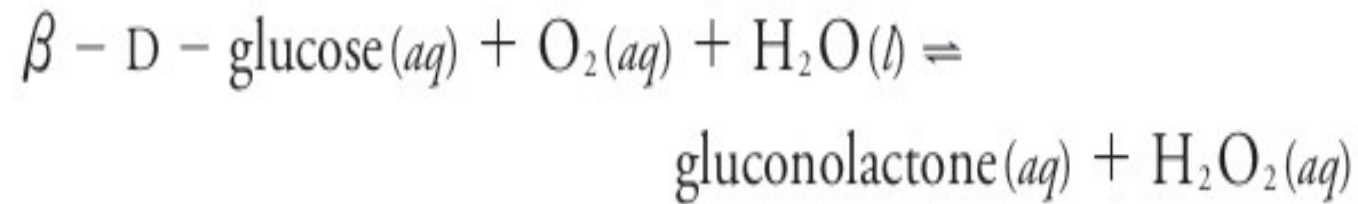
- Although several gases can diffuse across the gas permeable membrane (O_2 , N_2 , CO_2), only O_2 is reduced at the cathode.



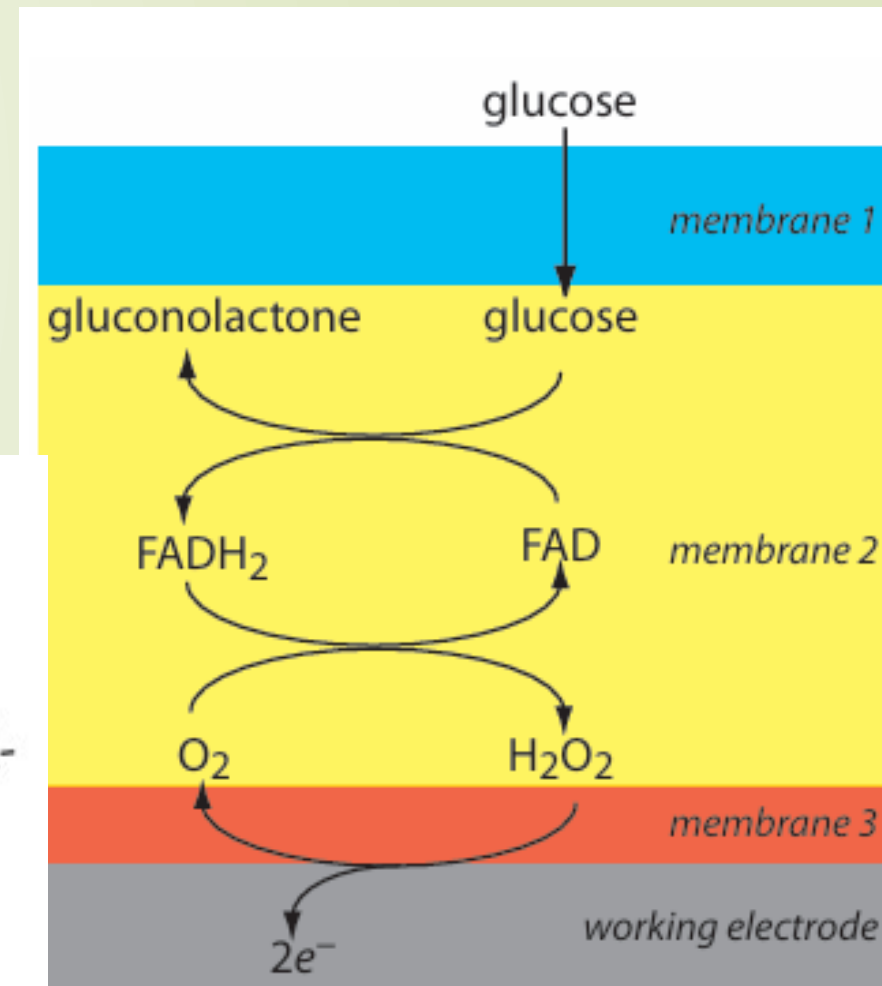
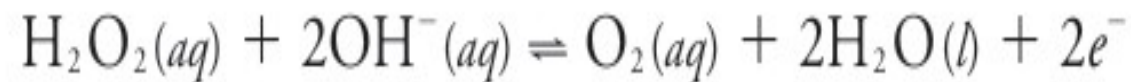
- Another e.g. amperometric biosensor of glucose
 - ✓ Biocatalytic membrane

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- The glucose oxidase enzyme catalyzes the oxidation of glucose.



The hydrogen peroxide diffuses through the innermost membrane of cellulose acetate where it undergoes oxidation at a Pt anode.



Amperometry titration

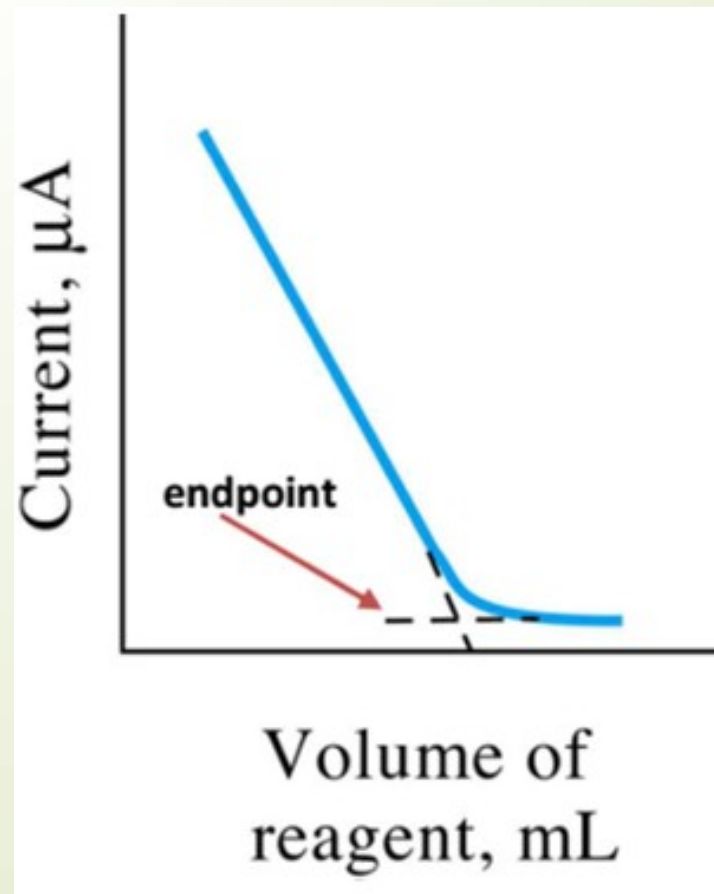
- **Locates the end point**
 - Based on the measurement of I produced by the titration reaction.
 - A plot of I as a fun of titrant volume
- The shape of the curve varies depending on:
 - The chemical species (titrant, analyte, products) that are electroactive & the revesribility of the redox couple.

- The plot consists of linear & nonlinear regions before & after the E.P & extrapolated to intersect at the end point.

A.Titration of electro-reducible ion with the nonreducible ion:

- In the first part of the plot, as reaction progresses the analyte's conc decreases, as does the diffusion current.
- At the end point, all the reducible are removed & show the constant diffusion current.

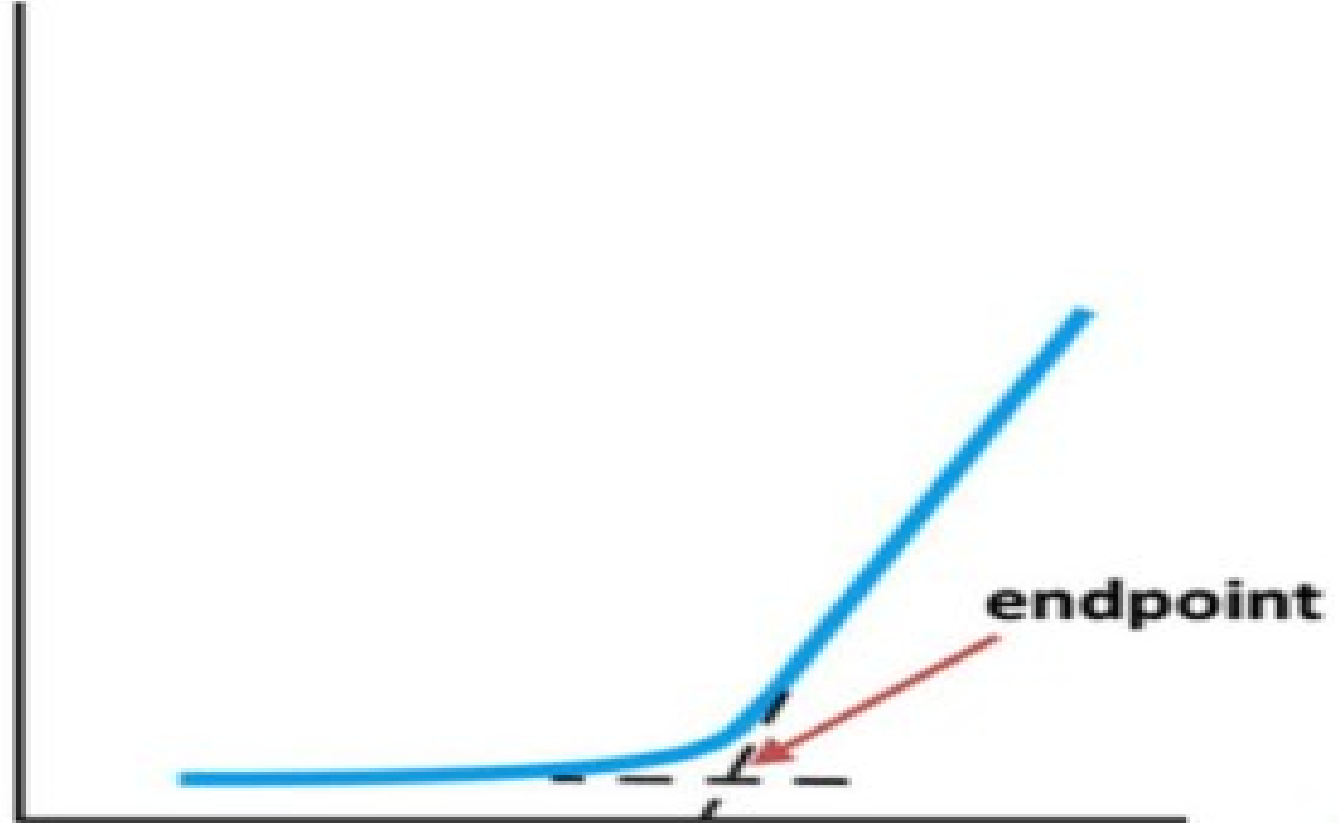
- E.g. Lead ions are titrated with the sulphate ions



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- Initially the diffusion current of the non-reducible ions is minimum.
- The addition of the titrant to the titrate does not show the change in the diffusion current.
- At the end point, the further addition of the titrant shows the gradual increase of the diffusion current.
- E.g. Titration of the chloride ions with the silver ions.

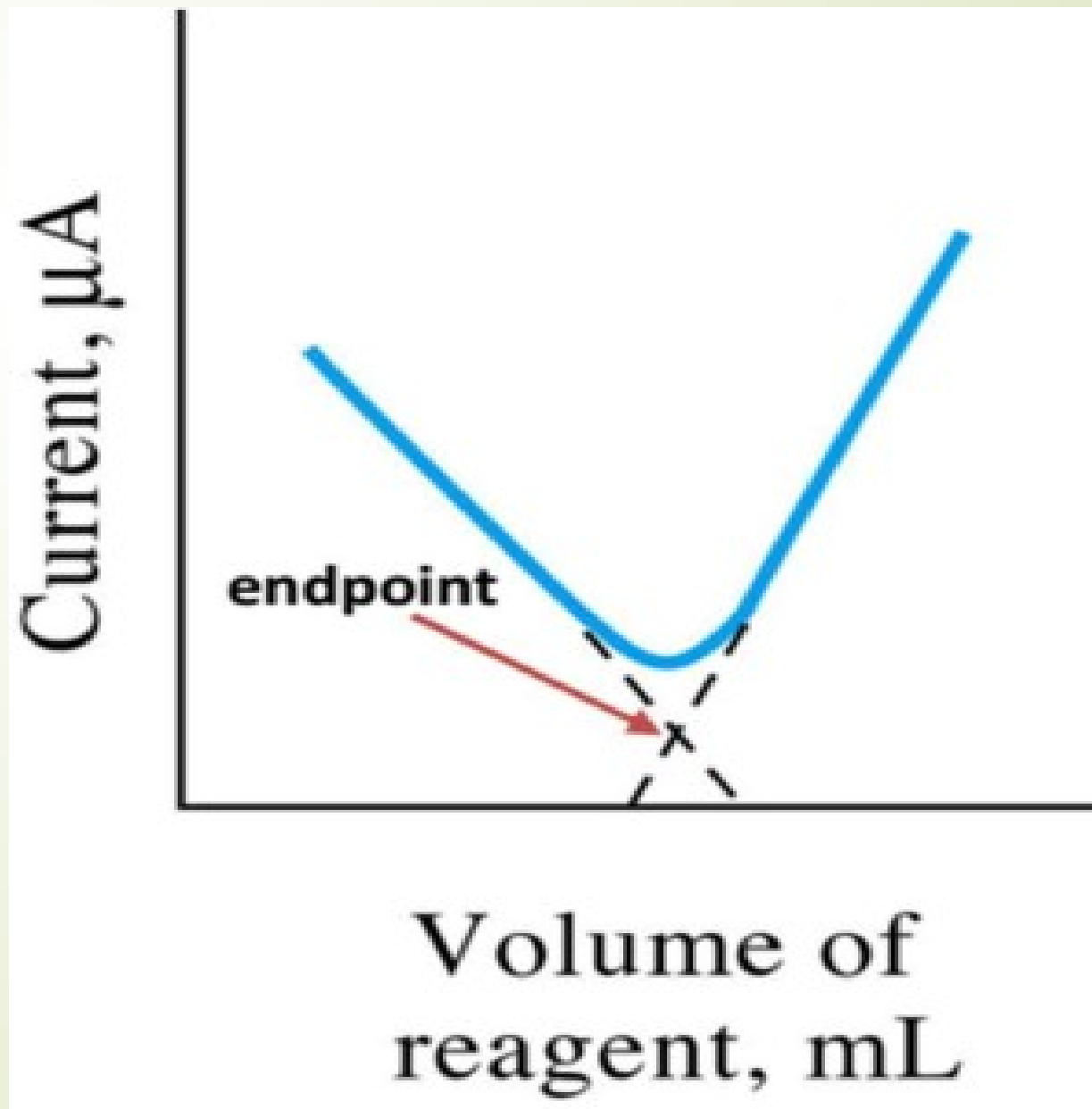
Current, μA



Volume of
reagent, mL

Titration of electro reducible ion with the electroreducible ions:

- The titration of the reducible ions with the reducible ions shows the decrease in the diffusion current as analyte's conc ↓se.
- Then the addition of the titrant increases the diffusion current as conc of the titrant ↑se.
- The curve shows the V-shaped curve.
- E.g. Titration of the lead ions with the dichromate ions.





Thank you